

Fig. 60.1 Baseline estimated glomerular filtration rate (eGFR) threshold below which the risk for end-stage kidney failure (ESKD) exceeded the risk of death for each age group. (From O'Hare AM, Choi AI, Berenthal D, et al. Age affects outcomes in chronic kidney disease. *J Am Soc Nephrol*. 2007;18:2758–2765.)

Fig_60_1

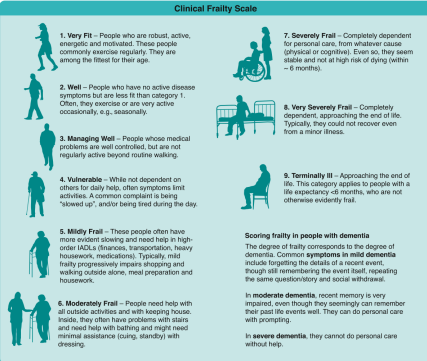


Fig. 60.2 Clinical frailty scale. ADLs, instrumental activities of daily living. (From Rockwood K, Song X, MacKnight C, et al. A global clinical measure of fitness and frailty in elderly people. *CMAJ*. 2005;173(8):489–495.)

Fig_60_2

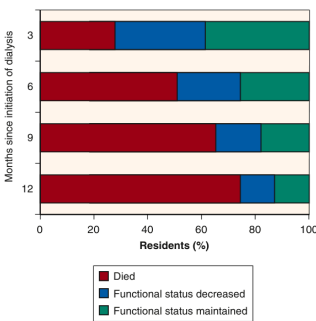


Fig. 60.3 Survival and functional status after the initiation of dialysis among nursing home residents. (From Kurella Tamura M, Covinsky KE, Chertow GM, et al. Functional status of elderly adults before and after initiation of dialysis. *N Engl J Med*. 2009;361:1539.)

Fig_60_3

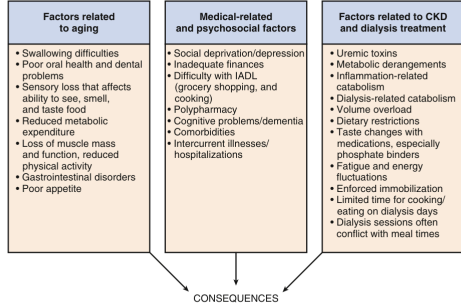


Fig. 60.4 Common factors related to aging and dialysis leading to malnutrition and protein-energy wasting (PEW). CKD, Chronic kidney disease; ADL, instrumental activities of daily living; QOL, quality of life.

Fig_60_4

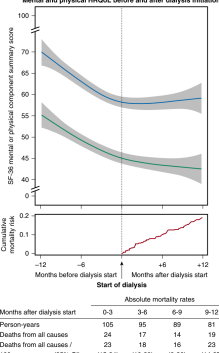


Fig. 60.5 Figure showing change in quality of life in older adults before, and after starting dialysis. (From de Rooij NM, Meuwissen Y, de Zeeuw D, et al. Quality of life before and after the start of dialysis in older patients. *Clin J Am Soc Nephrol*. 2012;17(8):1158–1167.)

Fig_60_5

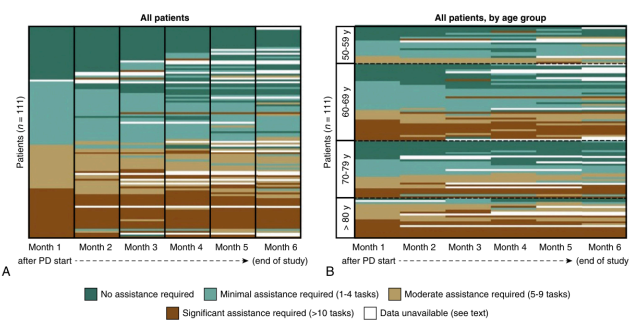


Fig. 60.6 (A and B) The change in need for assistance over time in older patients maintained on dialysis. The different shading patterns represent the degree of assistance provided. (From Fonseca-Correa J, Faragher JF, Tomlinson G, et al. Longitudinal changes in the use of PD assistance for patients maintained on peritoneal dialysis. *Kidney360*. 2021;2(3):469–476.)

Fig_60_6

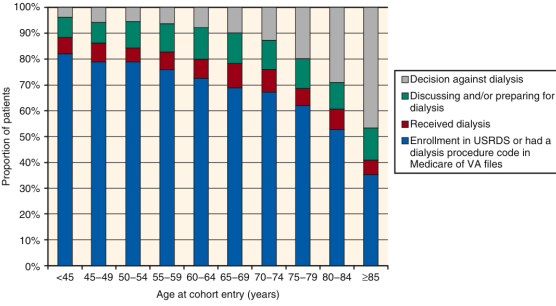


Fig. 60.7 Age differences in treatment decisions and practices for advanced kidney disease. Based on a chart review of a random sample of patients not enrolled in the U.S. Renal Data System and who did not have a dialysis procedure code in Medicare or Department of Veteran Affairs (VA) administrative files during follow-up. (From Wong SP, Hebert PL, Laundry RJ, et al. Decisions about kidney replacement therapy in patients with advanced kidney disease in the US Department of Veterans Affairs, 2000–2011. *Clin J Am Soc Nephrol*. 2016;11(10):1825–1833.)

Fig_60_7

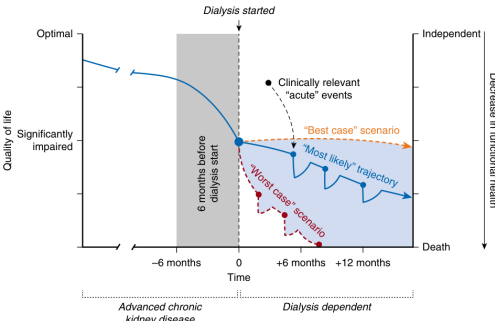


Fig. 60.8 Hypothetic life trajectory in older patients who start dialysis treatments across a spectrum of outcomes. (From Fonseca-Correa JI, Jassal SV. Health care for older adults with kidney failure. *Clin J Am Soc Nephrol*. 2022;17(8):1110–1112.)

Fig_60_8

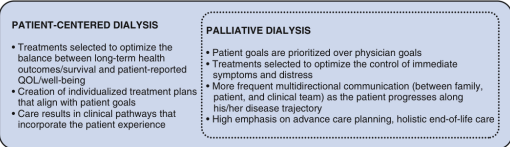


Fig. 60.9 Conceptualizing palliative dialysis. In patient-centered dialysis, treatments are chosen to align with patient goals. In palliative dialysis, a subset within patient-centered care, treatments are chosen so that if needed, patient goals are prioritized over physician goals. QOL, Quality of life.

Fig_60_9

60. Care of the Older Adult With Chronic Kidney Disease · assets 10-15 / 15

[illegible]

Table_60_1

[illegible]

Table_60_2

Risks	Details	References
Previous falls	One or more falls during the previous 12 mo	100,103,113
Fear of falling	Fearful anticipation of a fall; detectable on screening using the Falls Efficacy Scale (high scores)	107
Frailty	Predisposed to falls for multiple reasons (see text)	108,214
Balance and gait problems	Increased postural sway, failed walking test, can be exacerbated by arthritic changes or neuropathy	111,215,216
Pain	Pain may lead to guarding and abnormal posture due particularly to lower limb and foot pain	
Depression and antidepressants	Fall risk worsened by insomnia and sleep fragmentation, impaired attention and psychomotor slowing, and/or exacerbated using sedatives and antidepressants	313,315
Number of prescribed oral drugs	Increased fall risk, particularly with benzodiazepines, antidepressants, β -blockers, other antihypertensive agents, opiate derivatives	315,217,218
Cognitive impairment	Falls precipitated by decreased awareness of environmental risks; altered judgment and reduced executive functioning result in less effective compensation for age- or disease-associated changes in gait and balance	316,219
Sensory deficits	Poor vision and/or hearing affect environmental awareness, balance, and compensation strategies	
Diabetes	Impaired vision due to retinopathy; peripheral neuropathy, and hypoglycemic episodes increases the risk of falls	315,220
Anemia	May result in low energy, poor exercise tolerance, and postural hypotension	
Protein-energy wasting and anorexia	Loss of muscle mass and strength results in less effective strategies to compensate and maintain balance; reduced exercise capacity	313,221
HD therapy	Increased risk of transient or rapid fluid and electrolyte shifts, hypotension, arrhythmias; stiffness and muscle weakness exacerbated by forced mobility	322
PD therapy	Environmental risks—connected to a cyclor machine overnight, tubing around the bed may pose trip risk; mechanical factors—changes in center of gravity due to large volumes of intrabdominal fluid	108

Table_60_3

Drug Class	Safety, Dosing, and Efficacy of Commonly Administered Medications and Other Key Issues (Table 1) and Table 2	Concomitant	Ethnicity Issues	Clinical Adverse Effects
Sedative	Commonly used sedative in ICU (20%) Clonazepam 1 mg Lorazepam 1 mg Midazolam 1.5 mg Propofol 1 mg Thiopental 1 mg Valproic acid 1 mg Zolpidem 1 mg	Concomitant Benzodiazepines in ICU are common Lorazepam is published data in pediatric ICU Valproic acid is published data in pediatric ICU Zolpidem is published data in pediatric ICU	Ethnicity Issues The ethnic groups in ICU (African American, Hispanic, and Asian) are significantly higher than in the general population	Clinical Adverse Effects Sedation, respiratory depression, hypotension, hypoxemia, hypothermia, bradycardia, and prolonged QTc interval
Analgesic	Commonly used analgesic in ICU (20%) Morphine 1 mg Fentanyl 1 mg Hydromorphone 1 mg Ketorolac 1 mg Oxycodone 1 mg Propofol 1 mg Thiopental 1 mg Valproic acid 1 mg Zolpidem 1 mg	Concomitant Morphine is published data in pediatric ICU Fentanyl is published data in pediatric ICU Hydromorphone is published data in pediatric ICU Ketorolac is published data in pediatric ICU Oxycodone is published data in pediatric ICU Propofol is published data in pediatric ICU Thiopental is published data in pediatric ICU Valproic acid is published data in pediatric ICU Zolpidem is published data in pediatric ICU	Ethnicity Issues One study using ICU data showed that African American patients are more likely to receive opioids than white patients and are more likely to receive opioids than white patients	Clinical Adverse Effects Sedation, respiratory depression, hypotension, hypoxemia, hypothermia, bradycardia, and prolonged QTc interval
Cardiopulmonary	No drug adjustment No drug adjustment	Concomitant No drug adjustment	Ethnicity Issues One study using ICU data showed that African American patients are more likely to receive opioids than white patients and are more likely to receive opioids than white patients	Clinical Adverse Effects Sedation, respiratory depression, hypotension, hypoxemia, hypothermia, bradycardia, and prolonged QTc interval
Cardiovascular	No drug adjustment No drug adjustment	Concomitant No drug adjustment	Ethnicity Issues One study using ICU data showed that African American patients are more likely to receive opioids than white patients and are more likely to receive opioids than white patients	Clinical Adverse Effects Sedation, respiratory depression, hypotension, hypoxemia, hypothermia, bradycardia, and prolonged QTc interval
Neurological	No drug adjustment No drug adjustment	Concomitant No drug adjustment	Ethnicity Issues One study using ICU data showed that African American patients are more likely to receive opioids than white patients and are more likely to receive opioids than white patients	Clinical Adverse Effects Sedation, respiratory depression, hypotension, hypoxemia, hypothermia, bradycardia, and prolonged QTc interval

Table_60_4_part1

Table 60.4 Safety, Dosing, and Efficacy of Common Antidepressant Medications used in Chronic Kidney Disease (CKD) and End-Stage Kidney Disease (Continued)				
Drug Class, Generic	Dose Adjustment in CKD ¹⁴⁴	Comment	Efficacy Studies	Class Adverse Effects ^{144,145,146}
Venlafaxine	Normal dosage: 75–225 mg daily eGFR 10–70: consider reduction by 25%–50%. Dialysis: reduce total daily doses by 50%	5% excreted in urine. Clearance decrease and half-life prolonged in kidney impairment Manufacturer advises to avoid use if GFR <10 mL/min Accumulation of toxic metabolite can occur, myoglobinuria and kidney failure have been reported but rare	No efficacy data	Like SSRIs plus increased blood pressure Low toxicity seen with duloxetine
Duloxetine	No adjustment required if eGFR >30 Recommended starting at 30 mg every other day, titrating slowly to a max 60 mg daily; use not recommended with eGFR <30	No adjustment in patients with advanced CKD and on maintenance dialysis	No efficacy data	
Nonadrenergic and Specific Serotonergic Antidepressant				
Mirtazapine	No dosage adjustment recommended for eGFR >10 Start at 3.75 mg and up-titrate to effect; max 45 mg daily GFR <10: consider dose reduction	75% excreted unchanged in urine Clearance reduced by 50% in those with eGFR <10 mL/min Has been used to treat pruritus caused by kidney failure	No efficacy data	Appetite stimulation, weight gain, sedation
Norepinephrine-Dopamine Reuptake Inhibitor				

Table_60_4_part2

Risk Considerations	Benefit Considerations
Medication-Associated risk	
Is the medication cleared in whole or in part by the kidney?	What is the population in which this medication has been studied?
Does the medication have a narrow therapeutic window?	Does it include older adults or those with kidney disease? Is there observational or clinical trial evidence that benefits extend to those with kidney disease?
Is the medication thought to be of high risk in the general older population or in individuals with comorbidities, similar to those of the patient in question?	If patients with kidney disease have not been studied, does the medication have a track record of safety in postapproval studies?
Does the medication have potential central nervous system effects? Are data available to guide dosing in patients with kidney disease (e.g., pharmacokinetic studies, drug level monitoring)?	Would benefits of the medication be due to improvement in symptoms or in decrease in risk from asymptomatic disease? What are the patient's preferences in adding new medications?
Patient-associated risk	
Is the patient already taking multiple medications, poor vision, frailty?	Does the medication address a problem for which the patient is at significant risk (e.g., cardiovascular disease in patients with kidney disease)? Is the patient likely to accrue significant absolute risk reduction from the new medication?
Does the patient have cognitive dysfunction, poor vision, frailty?	
Risk of additional medications may be higher in this group.	
Does the patient have a history of adherence problems, and what would be the consequences of patient-related erratic dosing?	

From Rifkin DE, Winkelmayer WC. Medication issues in older individuals with CKD. *Adv Chronic Kidney Dis*. 2010;17:320–328.

Table_60_5